Practical No.4 Simple Linear Regression using R

Complete Simple R Program

```
# Create data
data <- data.frame(
  Hours = c(1, 2, 3, 4, 5, 6),
  Marks = c(40, 45, 50, 60, 65, 75)
)
```

Output:

```
Hours Marks
1     1    40
2     2    45
3     3    50
4     4    60
5     5    65
6     6    75
```

```
# Create regression model
model <- lm(Marks ~ Hours, data = data)
```

Output:

```
Coefficients:
(Intercept)       Hours
      31.33        7.00
```

```
# Display coefficients
coef(model)
```

Output:

```
coef(model)
(Intercept)       Hours
   31.33333     7.00000
```

```
# Predict marks for 6 hours
predict(model, data.frame(Hours = 6))

# Plot data
plot(data$Hours, data$Marks,
     main = "Simple Linear Regression",
     xlab = "Hours Studied",
     ylab = "Marks")

# Add regression line
abline(model)
```

Output:

